

MOHAMMED JAWWAAD SHERIFF

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SUMMARY

Final-year Computer Science undergraduate whose strongest work is taking an existing implementation apart and measuring where it actually goes wrong. Recent work: auditing a team ML pipeline and proving the trained network emitted a byte-identical output vector for every input, and fine-tuning a 300M-parameter speech model to a measured 34.6% WER inside a hard 12-hour GPU budget. Python, Java, C++, SQL and TypeScript; used to working on code other people wrote.

EDUCATION

PES University, B.Tech in Computer Science & Engineering
Bengaluru, Karnataka, India | Active Backlogs: 0

Aug 2023 – Jun 2027 (Expected)

TECHNICAL SKILLS

Programming: Python, Java, C, C++, SQL, JavaScript (ES6+), TypeScript

CS Fundamentals: Data Structures & Algorithms, Time & Space Complexity Analysis, Operating Systems, Computer Networks, DBMS, OOP & Design Patterns

Analysis & Optimisation: Instrumentation and measurement, baseline comparison, blind-test validation, cross-validation, deduplication and record linkage, compute and memory budgeting on constrained GPUs

Backend & Data: Node.js, FastAPI, Flask, REST APIs, PostgreSQL, MySQL, MongoDB, Prisma, SQLAlchemy, SQLite

ML & AI: PyTorch, Hugging Face Transformers, Scikit-learn, XGBoost, Pandas, NumPy, RAG, LangChain, AI Agents

Frontend & Tools: React.js, Next.js, Git, GitHub, Docker, Linux, Vercel, Agile

PROJECTS

Multilingual Speech Recognition – Urdu ASR (Wav2Vec2 XLS-R)

2026

PyTorch, Hugging Face Transformers, CTC, Common Voice 17, Kaggle GPU

- Fine-tuned the 300M-parameter wav2vec2-xls-r-300m with a CTC head on Common Voice 17 Urdu, reaching 34.6% WER and 11.5% CER over the full 4,056-clip test set rather than a sample of it. Published the model and a Gradio demo so the number is reproducible by anyone.
- Ran the whole thing inside a hard 12-hour free-GPU session cap: traced where the budget was actually going and found full-test-set evaluation every 400 steps was eating most of it, and that the platform's P100 was unsupported by the installed PyTorch build — moving to dual T4s got the run finishing.

Wildfire Risk Pipeline – Validation and Correctness Audit

Final-Year Capstone, 2026

Python, PyTorch, XGBoost, NumPy, Pandas, rasterio

- Owned blind-test validation on a four-person capstone. Ran the saved model artifacts as a black box and found the trained CNN branch returned a byte-identical output vector for every input, and that the late-fusion head was a monotone transform of the XGBoost branch (Spearman = 1.000).
- Traced the cause down into the data layer: 94% of samples were being fed all-zero patches because only 4 of 54 grid cells fell inside the raster bounds, and the vegetation raster was being indexed with the slope raster's pixel index at a different resolution.
- Found a left join duplicating weather rows once per satellite detection, which lifted the positive rate from 20% to 56% and inflated the headline accuracy. Re-measured on de-duplicated data: 89.9% against an 85.0% majority baseline, ROC-AUC 0.907 and PR-AUC 0.686.

ASCEND – Deployed Productivity Platform

2026

Next.js 16, React 19, TypeScript, Prisma, PostgreSQL, NextAuth, Vercel

- Designed an 11-model relational schema and built the whole application on it — planner, calendar, workout logger, focus timer, habit streaks, friends and analytics — deployed on Vercel and in daily use by a small group of real users.
- Ran a security pass over my own API before opening it up: auth on every route, cross-user IDOR returning 404s, parameterised queries, schema validation stripping unknown fields, bcrypt hashing, HttpOnly/Secure/SameSite cookies, security headers and per-route rate limiting.

Satellite Ground Station Automation

2026

Python, Skyfield, SatNOGS REST API, Hamlib rotctld

- Wrote the tracking engine: computes azimuth and elevation from live TLEs with Skyfield and drives a Hamlib rotator, with a hard 25° elevation floor so the dish cannot tip under its own weight.
- Matched 21 requested satellites against inconsistent naming in the SatNOGS database, dropped duplicates, and tested the full loop against a rotator simulator I wrote, because the hardware did not exist yet.

Additional Projects: Job Hunt Command Center (FastAPI, SQLAlchemy, APScheduler): scheduled crawler over 166 public ATS boards, ~13k postings a run, with dedupe rules that still keep genuinely distinct postings apart. • Crime Risk Prediction (Scikit-learn, Streamlit): reconciled three public datasets with conflicting state names down to 28 clean records, Random Forest with stratified 3-fold cross-validation. • MediBot (ROS2 Humble, Nav2, SLAM Toolbox): autonomous hospital-floor navigation across 6 rooms with a priority-aware delivery queue.

EXPERIENCE

Software Development Intern, Ryzklytix Consulting Solutions Pvt Ltd

Jul 2025 – Aug 2025

Bengaluru, India

- Built client-facing features in React and Node at a consulting firm, shipping against real delivery deadlines.
- Paired with senior engineers on REST API integration and component structure, working through code review, sprint planning and deployment cycles.

CERTIFICATIONS, ACHIEVEMENTS & LEADERSHIP

Certifications: Problem Solving (Intermediate), HackerRank (2024) • Deep Learning Essentials – Neural Networks with PyTorch (2024)

Achievements: C-HEAL Hackathon Shortlisted Finalist (2024), picked out of a campus-wide field to pitch a healthcare prototype to industry mentors. Also took part in HealthHack 3.0.

Leadership: Founded PES University's first Clash Royale campus tournament (2024), owning logistics, scheduling and live operations; Core Logistics, Samarpana '24; S&I Volunteer, TAMS. State-level swimmer and former School Swimming Team Captain.